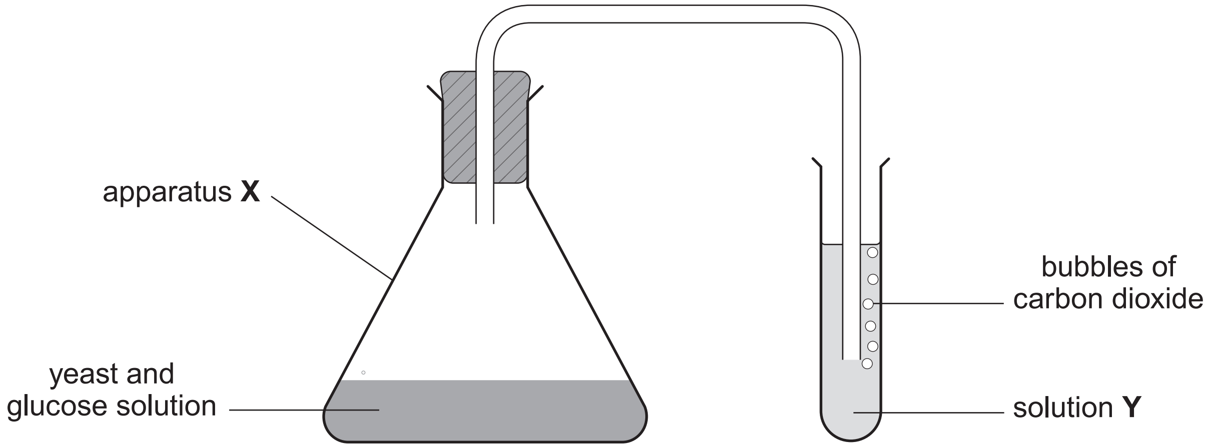
Examiner only

3. (a) Ethanol can be produced by fermentation using the apparatus shown.



(i) Circle the name of apparatus **X**. [1]

conical flask **measuring cylinder** **beaker**

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(ii) Solution **Y** can be used to show the presence of carbon dioxide gas.

Give the name of solution **Y** and state what is seen when carbon dioxide is bubbled through it. [2]

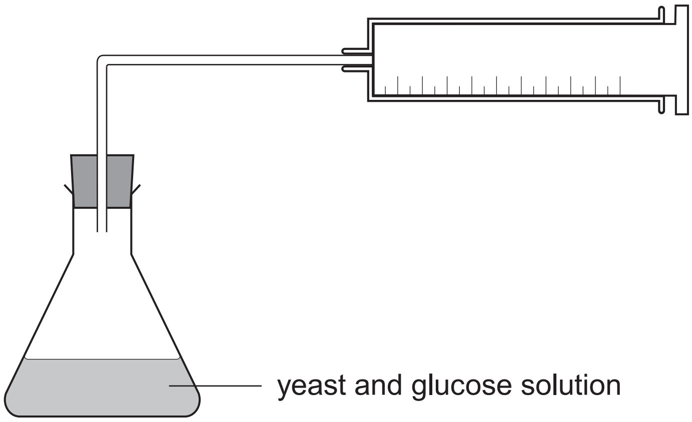
Name of solution **Y** ....

What is seen ....

(iii) The chosen temperature for this reaction is 35 °C. Tick (✓) the correct reason why carbon dioxide would **not** be produced at a temperature of 90 °C. [1]

the reaction is finished	
the yeast is used up	
the enzymes in the yeast are denatured	

(b) In a different experiment, using the apparatus shown below, the volume of carbon dioxide produced at 35 °C was measured and recorded every 10 minutes for 60 minutes.



Time (minutes)	Volume of carbon dioxide produced (cm^3)
0	0
10	6
20	12
30	17
40	25
50	30
60	36

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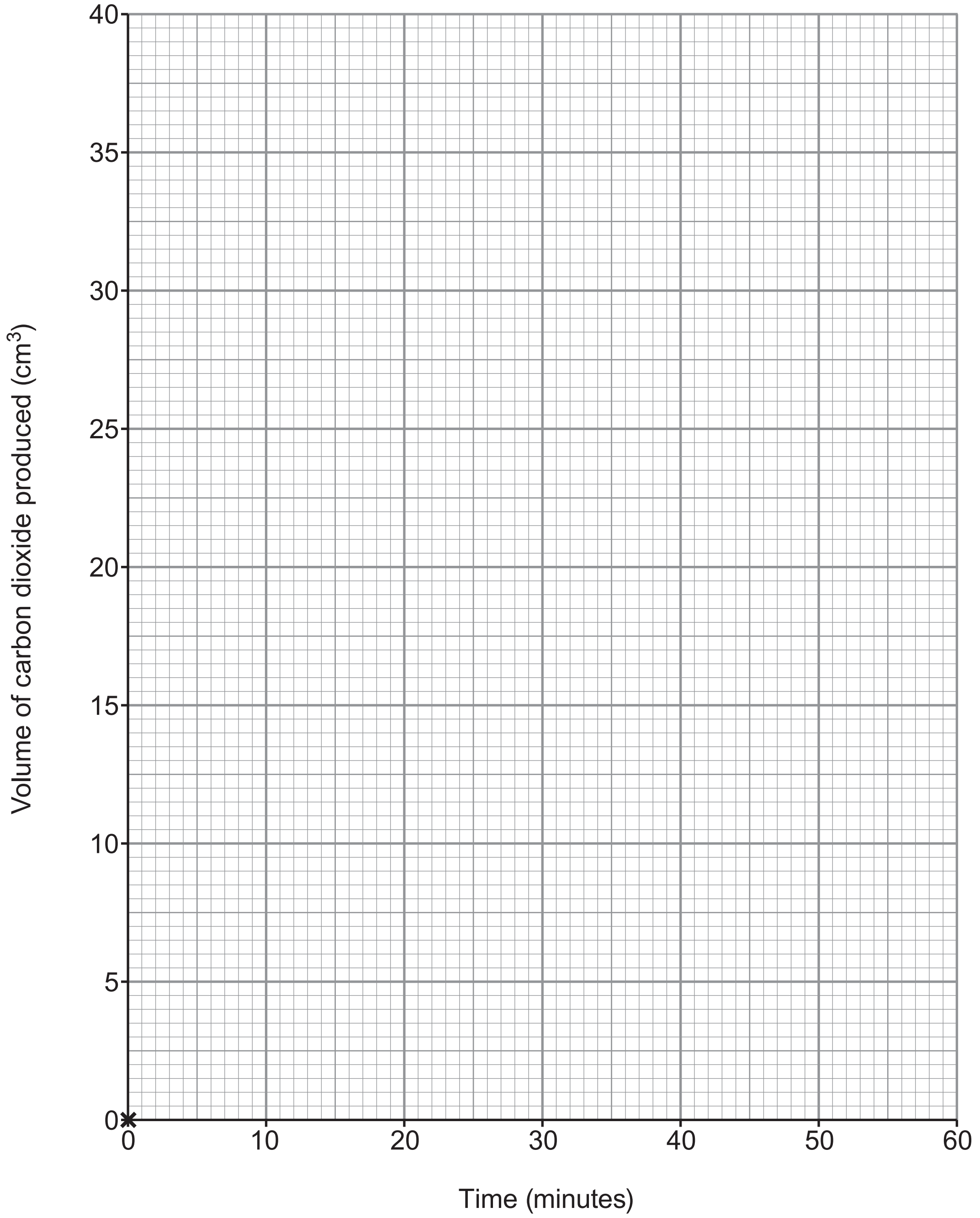
(i) Plot the data on the grid below and draw a suitable line.

The first point has been plotted for you. [3]



Examiner only

(ii) Use the graph to find the volume of carbon dioxide produced after 25 minutes. [1]

Volume .... cm^3

(iii) Use the data to **estimate** how long it would take to produce $100\,cm^3$ of carbon dioxide gas. Assume that the rate of the reaction does not change. [1]

Time = ... minutes

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